Customer Shopping Behavior Analysis Report

A Data Analyst Portfolio Project

Exploring consumer patterns through SQL analysis and interactive visualization

Author Database Dataset Tools Rajan Nanda PostgreSQL 18 699 Rows · 18 Cols SQL ·

Python · Dash

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1. Project Overview & Objectives

This project is a comprehensive data analysis of customer shopping behavior using a structured relational dataset stored in PostgreSQL. The goal is to extract actionable business intelligence by answering 10 targeted analytical questions using SQL, then visualizing the results through an interactive Python Dash dashboard.

This report demonstrates the complete data analyst workflow: from understanding the dataset and formulating business questions, to writing production-quality SQL queries, interpreting results, and communicating findings clearly through visualizations and written analysis.

Project Objectives

• Understand revenue distribution across customer demographics (gender, age group)
• Identify the impact of discounts and subscriptions on customer spending behaviour •
Determine top-performing products by sales volume and customer review ratings •
Segment customers by loyalty level to support targeted marketing strategies
• Analyse shipping preferences in relation to purchase amounts
• Build an interactive dashboard to present findings to non-technical stakeholders

699 Total Customers	**18** Total Columns	**10** SQL Queries	**11** Visualizations

2. Dataset Description

The dataset, named **customer_shopping_behavior**, contains 699 records of individual customer transactions captured across various shopping categories in the United States. Each record represents a single purchase event with full demographic, product, payment, and satisfaction details.

Dataset Schema — Key Columns

customer_id	INTEGER	Unique customer identifier
age	INTEGER	Customer age in years
age_group	TEXT	Derived: Young Adult / Adult / Middle-aged / Senior
gender	TEXT	Male or Female
item_purchased	TEXT	Name of the product purchased
category	TEXT	Clothing / Accessories / Footwear / Outerwear
purchase_amount	NUMERIC	Transaction value in USD
location	TEXT	US State where purchase was made
size	TEXT	Product size (S / M / L / XL)
color	TEXT	Product color
season	TEXT	Season of purchase (Spring/Summer/Fall/Winter)
review_rating	NUMERIC	Customer review score (1.0 – 5.0)
subscription_status	TEXT	Whether customer has active subscription (Yes/No)
shipping_type	TEXT	Shipping method selected
discount_applied	TEXT	Whether a discount was used (Yes/No)
promo_code_used	TEXT	Whether a promo code was applied
previous_purchases	INTEGER	Number of past purchases by this customer
payment_method	TEXT	Payment method used

3. Tools & Technologies

PostgreSQL	18	Relational database engine — data storage and SQL analysis
pgAdmin 4	Latest	GUI for writing and executing SQL queries
Python	3.14	Dashboard development and database connectivity
psycopg2	2.9+	PostgreSQL adapter for Python
Pandas	2.x	Data manipulation and query result handling
Plotly	6.x	Interactive chart and visualization library
Dash	4.x	Web-based interactive dashboard framework
ReportLab	4.x	PDF report generation

4. Methodology

This project follows the standard Data Analyst workflow — from data understanding through to communication of insights:

→

Step 1 — Data Exploration →

Step 2 — Question Formulation

→

Step 3 — SQL Query Development

→

Step 4 — Result Validation →

Step 5 — Visualization

→

Step 6 — Reporting

Examined the dataset schema, identified column types, checked for null values (e.g. review_rating had nulls), and understood the domain context of customer shopping data.

Defined 10 business questions spanning revenue analysis, customer segmentation, product performance, discount behaviour, and logistics — covering all major analytical dimensions of the dataset.

Wrote production-quality SQL queries using aggregations, subqueries, CASE expressions, Common Table Expressions (CTEs), and window functions (ROW_NUMBER with PARTITION BY) in PostgreSQL.

Executed all queries in pgAdmin, verified result counts and values for logical consistency, and cross-checked edge cases (e.g. null ratings cast to numeric before aggregation).

Connected Python to PostgreSQL via psycopg2, ran all queries programmatically using pandas.read_sql_query(), and built 11 interactive

Plotly charts within a structured Dash dashboard. insights into this formal project report for portfolio presentation.

Compiled all queries, results, findings, and

5. Analysis & Findings — 10 Business Questions

What is the total revenue generated by male vs. female customers?

SQL Query

```
SELECT gender, SUM(purchase_amount) AS revenue

FROM customer

GROUP BY gender;
```

Results

Male	$~130,000+
Female	$~102,000+

■ Male customers generate higher total revenue than female customers. However, this may reflect a larger proportion of male customers in the dataset rather than higher individual spending. Average spend per customer should be compared for a more meaningful conclusion.

Which customers used a discount but still spent more than the average purchase amount?

SQL Query

```
SELECT customer_id, purchase_amount

FROM customer

WHERE discount_applied = 'Yes'

AND purchase_amount > (SELECT AVG(purchase_amount) FROM customer);
```

Results

Customers matching criteria	~180+ customers
Subquery — Dataset avg spend	~$59.77 USD
Min spend in this group	> $59.77 USD

■ A significant portion of discount users are high-value customers. This challenges the assumption that discounts only attract low-spend buyers. These customers represent a strong target segment — they respond to promotions yet maintain above-average spend.

Which are the top 5 products with the highest average review rating?

SQL Query

```
SELECT item_purchased,

ROUND(AVG(review_rating::numeric), 2) AS avg_rating

FROM customer

GROUP BY item_purchased

ORDER BY avg_rating DESC

LIMIT 5;
```

Note: review_rating::numeric cast required because column contains null values.

■ The top-rated products maintain average ratings above 4.0/5.0, indicating strong customer satisfaction in specific product lines. These products are candidates for increased promotional spend and featured placement.

Compare average purchase amounts between Standard and Express Shipping.

SQL Query

```
SELECT shipping_type, ROUND(AVG(purchase_amount), 2) AS

avg_purchase FROM customer

WHERE shipping_type IN ('Express', 'Standard')

GROUP BY shipping_type;
```

■ Express and Standard shipping customers show similar average spend levels, suggesting that shipping preference is driven by urgency rather than budget. Express shipping customers may be higher-priority targets for loyalty programs given their time-sensitivity.

Do subscribed customers spend more? Compare average spend and total revenue.

SQL Query

```
SELECT subscription_status,

COUNT(customer_id) AS Total_Customer,

ROUND(AVG(purchase_amount),2) AS Avg_spend,

SUM(purchase_amount) AS Total_Revenue

FROM customer

GROUP BY subscription_status

ORDER BY Total_Revenue, Avg_spend DESC;
```

■ Subscribed customers contribute significantly to total revenue. Even if average spend per transaction is similar, the volume of subscription customers drives disproportionate total revenue. Growing the subscriber base should be a key business priority.

Which 5 products have the highest percentage of purchases with discounts applied?

SQL Query

```
SELECT item_purchased,

ROUND(100 * SUM(CASE WHEN discount_applied = 'Yes'

THEN 1 ELSE 0 END) / COUNT(*), 2) AS discount_rate

FROM customer

GROUP BY item_purchased

ORDER BY discount_rate DESC
```

```
LIMIT 5;
```

■ Some products show discount application rates above 50%, meaning more than half of their purchases involved a discount. This may indicate price sensitivity in those product lines, or overly aggressive discount strategies that could be eroding margin unnecessarily.

Segment customers into New, Returning, and Loyal based on purchase history.

SQL Query

```
WITH Customer_type AS (

SELECT customer_id, previous_purchases,

CASE

WHEN previous_purchases = 1 THEN 'New'

WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'

ELSE 'Loyal'

END AS customer_segment

FROM customer

)

SELECT customer_segment, COUNT(*) AS "Number of Customers"

FROM customer_type

GROUP BY customer_segment;
```

Technique Used

CTE (WITH clause)	Isolate customer classification logic before aggregating
CASE expression	Classify each customer into a segment based on purchase count
GROUP BY + COUNT(*)	Count how many customers fall into each segment

■ The majority of customers fall into the Returning segment, indicating a healthy retention rate. The Loyal segment (10+ purchases) represents the highest-value cohort and should receive priority treatment through VIP programs and early access to new products.

What are the top 3 most purchased products within each category?

SQL Query

```
WITH item_counts AS (

SELECT category, item_purchased,

COUNT(customer_id) AS total_orders,

ROW_NUMBER() OVER(

PARTITION BY category

ORDER BY COUNT(customer_id) DESC

) AS item_rank

FROM customer

GROUP BY category, item_purchased

)

SELECT category, item_purchased, total_orders

FROM item_counts

WHERE item_rank <= 3;
```

Technique Used

Window Function ROW_NUMBER()	Rank products within each category independently
PARTITION BY category	Restart ranking for every category
CTE (WITH clause)	Separate ranking logic from final filter
WHERE item_rank <= 3	Extract only top 3 per category

■ Window functions allow category-level ranking without complex self-joins. This is the most technically advanced query in the project and demonstrates proficiency with analytical SQL. Each category has clear top performers that should be prioritized in inventory and marketing.

Are customers with more than 5 previous purchases likely to subscribe?

SQL Query

```
SELECT subscription_status,

COUNT(customer_id) AS Repeat_buyers

FROM customer

WHERE previous_purchases > 5

GROUP BY subscription_status;
```

■ Among repeat buyers with more than 5 previous purchases, subscription rates are notably higher — confirming that loyalty and subscription status are positively correlated. Targeting repeat buyers with subscription offers is a high-conversion opportunity.

What is the revenue contribution of each age group?

SQL Query

```
SELECT age_group, SUM(purchase_amount) AS total_revenue

FROM customer

GROUP BY age_group

ORDER BY total_revenue DESC;
```

Young Adult	$62,143	#1 — Highest
Middle-aged	$59,197	#2
Adult	$55,978	#3
Senior	$55,763	#4 — Lowest

■ Young Adults are the top revenue-generating age group, followed closely by Middle-aged customers. Seniors, despite potentially higher disposable income, contribute the least — suggesting potential for targeted outreach or senior-friendly product lines.

6. Dashboard & Visualization

An interactive dashboard was built using Python Dash and Plotly, connecting directly to the PostgreSQL database via psycopg2. All 10 queries run automatically on startup and populate 11 dynamic charts organized into 4 analytical sections.

Dashboard Architecture

Data Source	PostgreSQL	my_practice_db · customer table · 699 rows
DB Connector	psycopg2	Direct connection from Python to PostgreSQL
Data Layer	Pandas	pd.read_sql_query() for each of the 10 queries
Charts	Plotly	Bar, histogram, donut, horizontal bar, dual-axis
Web Framework	Dash	Flask-based local web server at localhost:8050
Report	ReportLab	This PDF generated programmatically

Dashboard Sections

■ Revenue & Customer Overview	Q1 · Q7 · Q10 — Gender revenue, loyalty segments, age group revenue
■ Subscriptions, Discounts & Loyalty	Q5 · Q9 · Q2 — Subscriber value, repeat buyer patterns, high-spend discount users
■ Product Ratings & Discounts	Q3 · Q6 — Top-rated products, discount-heavy products
■ Category Leaders & Shipping	Q8 · Q4 — Best-sellers per category, shipping type analysis

To launch the dashboard, run:

```
python3 customer_dashboard.py
```

```
# Then open: http://127.0.0.1:8050
```

7. Key Business Insights

Male customers generate higher total revenue, but age group analysis shows Young Adults are the strongest demographic segment overall.
Most customers are in the 'Returning' segment, confirming good retention. Loyal customers (10+ purchases) represent the highest-value cohort and should be prioritized.
Discounts are not only attracting low-spend customers — a large group of discount users still spend above average. Discounting strategy needs refinement to protect margins.
Subscribed customers drive disproportionate total revenue. Subscription conversion of repeat buyers should be a top growth strategy.
Certain products consistently receive high ratings AND high discount rates — these may be candidates for premium pricing experiments.
Express and Standard shipping customers show similar spending levels, suggesting shipping choice is a logistics preference rather than a spend indicator.

8. Conclusion & Recommendations

This project successfully demonstrates a complete end-to-end data analysis workflow applied to a real-world retail customer dataset. Using PostgreSQL and advanced SQL techniques including CTEs, subqueries, CASE expressions, and window functions, 10 meaningful business questions were answered with data-driven precision.

The resulting insights provide clear, actionable direction across four business dimensions: customer segmentation, product strategy, discount optimization, and subscription growth. The interactive Python Dash dashboard makes these findings accessible to both technical and non-technical stakeholders.

Recommendations

→ **Invest in Young Adults**

Target marketing and product development toward the Young Adult segment, which generates the highest revenue. Personalized digital campaigns will be most effective for this demographic.

→ **Convert Repeat Buyers**

Implement an automatic subscription offer trigger after 5 purchases. Data shows repeat buyers have significantly higher subscription rates — this is a high-conversion touchpoint.

→ **Refine Discount Strategy**

Audit discount application rates by product. Products with 50%+ discount rates may be over-discounted. Test price sensitivity by reducing discounts and measuring impact on conversion.

→ **Reward Loyal Customers**

Launch a VIP program for Loyal segment customers (10+ purchases). Priority access, free express shipping, and exclusive products will reinforce loyalty and increase lifetime value.

→ **Promote Top-Rated Products**

Feature the highest-rated products prominently in campaigns and on-site placement. High ratings are social proof — leverage them to drive conversion in lower-performing lines.

→ **Expand Dataset for Depth**

Future analysis should include date/time of purchase to enable seasonal trend analysis, cohort retention curves, and time-series revenue forecasting.